



BUSE37802
Pricing Strategies

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UNDERSTANDING AND USING THE OWN-PRICE-ELASTICITY OF DEMAND

Objectives

- Define price elasticity.
- Understand how demand elasticity varies along a demand curve.
- Understand how to set prices given the demand elasticity.

Introduction

In the EVC reading, we derived demand curves. In this note, we want to understand how managers represent differences in these demand curves. For example, in the previous note we derived two different demand curves. How might we summarize these differences to a manager?

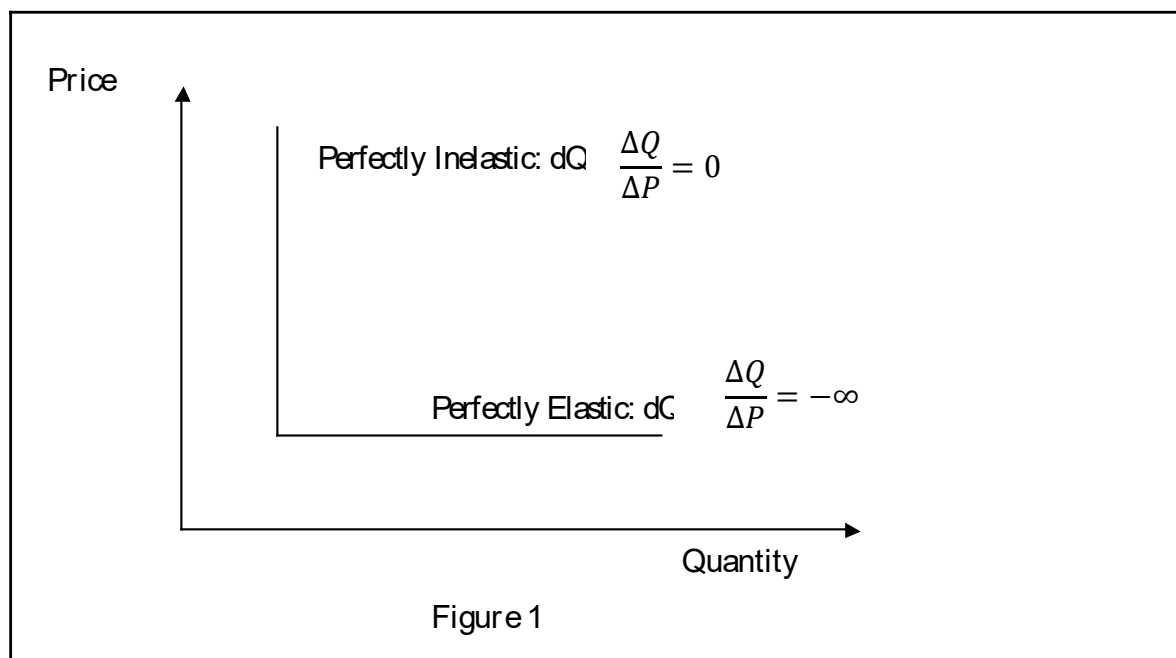
The most commonly used measure to summarize this information is price elasticity. A price elasticity is a unit-less measure that defines how a percent change in price leads to a percent change in demand. Mathematically, the price elasticity of demand is defined as follows:

$$\beta = \frac{\% \text{Change in Quantity}}{\% \text{Change in Price}} = \frac{P}{Q} \frac{\Delta Q}{\Delta P}$$

where P is the current price, Q is the current quantity, and $\frac{\Delta Q}{\Delta P}$ is the slope of the demand function $Q(P)$. As we normally use P as the y-axis and Q as the x-axis, $\frac{\Delta Q}{\Delta P}$ is the inverse slope. In words, it is often simplest to think of an elasticity as the percent change in quantity associated with a 1% change in price. Hence, if someone tells you that the elasticity is -3, then a 1% decrease in price will lead to a 3% increase in sales.

Extreme Cases

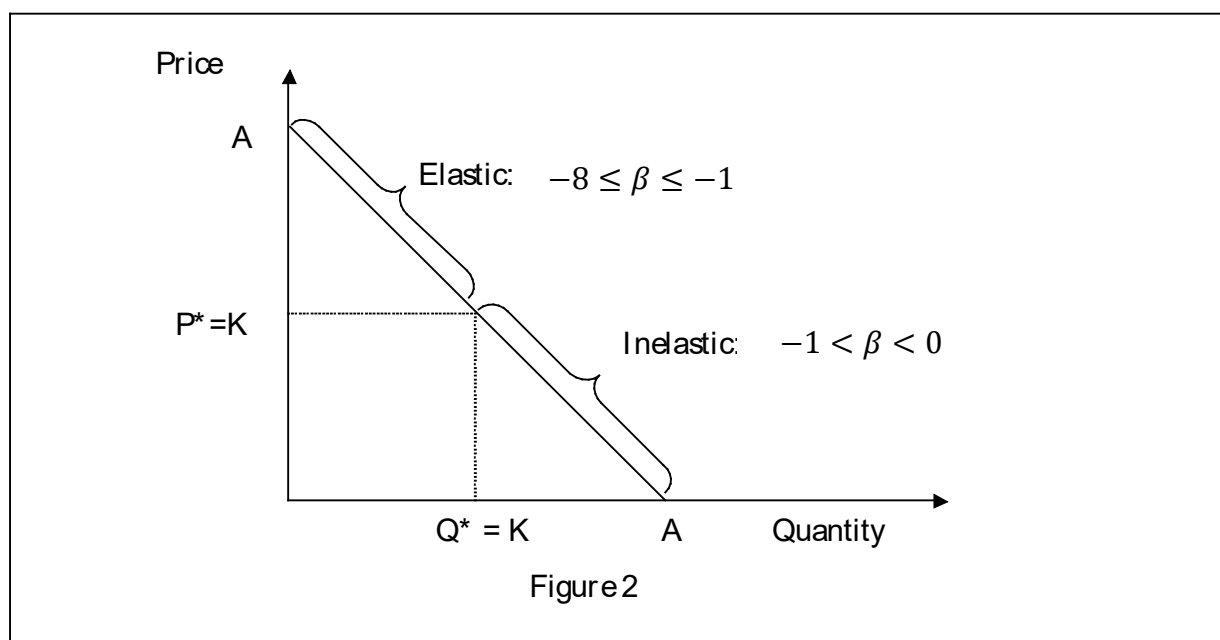
In the extreme case of a vertical demand curve, $\beta = 0$, we use the term perfectly *inelastic* demand. In the extreme case of a horizontal demand curve, $\beta = -\infty$, we use the term perfectly *elastic* demand. We tend to refer to “flat” demand curves as “more elastic” and “steep” demand curves as “more inelastic.”



Elasticity and Demand Curves

Elasticity Along a Demand Curve

For most demand curves, the elasticity changes as one moves along the demand curve. For example, consider the case of linear demand with a slope of -1 ($\frac{\Delta P}{\Delta Q} = -1$). At the point where $P=Q \equiv K$, $\beta = -1$ (unit elasticity). At the point $(K/2, 3K/2)$, $\beta = -1/3$ (inelastic). At the point $(3K/2, K/2)$, $\beta = -3$ (elastic).

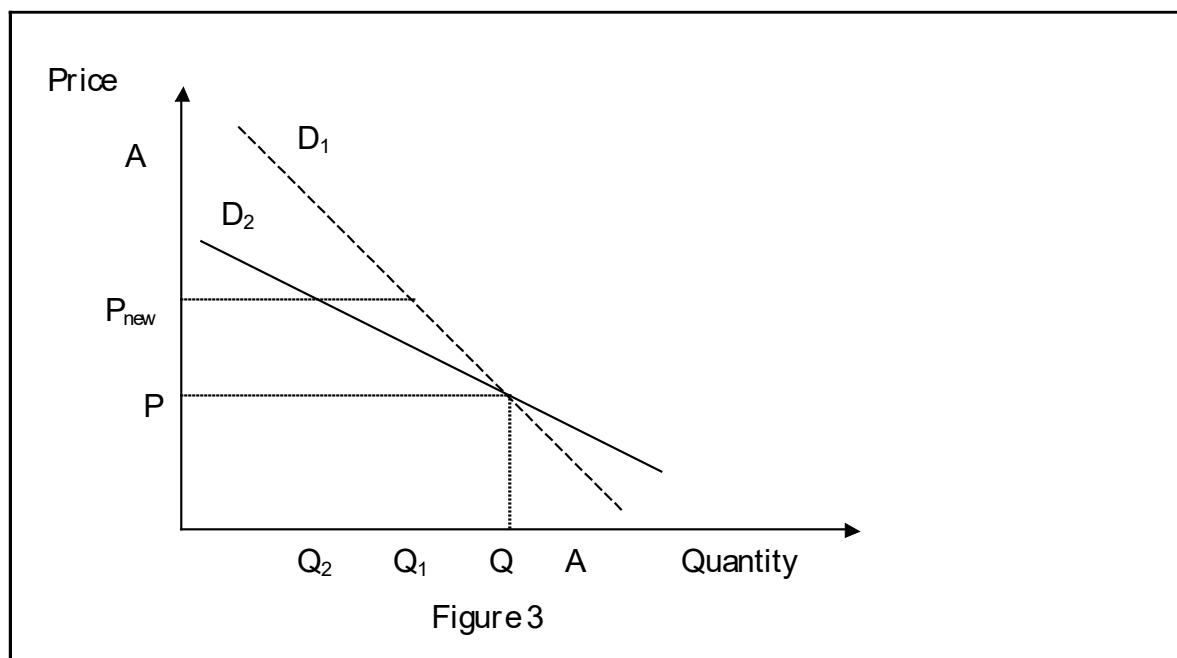


Notice that for $Q > Q^*$, demand is inelastic; for $Q < Q^*$ demand is elastic. In the formula for elasticity, notice that the slope in this example is constant. What drives changes in elasticity is changes in P and Q as we move along the demand curve. As one lowers the price, demand becomes more inelastic. Note that this is true for any downward sloping demand curve. The ratio of Price to Quantity approaches zero as price decreases to zero for any demand curve. Our example is special in that the slope is constant.

What you should take away from Figure 2 is that elasticity depends on both the slope and the magnitude of P and Q . Even if the slope is held constant, the elasticity along the demand curve can change due to changes in P and Q .

Elasticity of Different Demand Curves

Consider the two demand curves in Figure 3. One we would argue that demand curve D_2 is “flatter” than D_1 . That is, D_2 is more elastic. For a price increase of the same magnitude, P to P_{new} , there was a greater decrease in quantity demanded for D_2 . In most of our analysis, we want to hold P and Q fixed and then consider variations in the slope around this point. In this example, we would conclude that D_2 is more elastic than D_1 .



Pricing and Elasticity

Monopoly Pricing and Elasticity

The monopoly markup rule is given by:

$$\frac{Price - Cost}{Price} = -\frac{1}{\beta}$$

Monopoly Markup Rule

This rule characterizes the price level that maximizes profits. In other words, price is adjusted until the contribution margin divided by the price equals minus the inverse of elasticity. This formula is derived from maximizing monopoly profits using calculus. In practice, consultants and academics have shown that measured elasticities for branded consumer packaged goods are typically between -2 and -3 (see Tellis, 1988 and Dolan and Simon for examples).

Another fact we know about monopolists is that they will never produce along the inelastic region of the demand curve. This is easy to show:

The rule tells us that if marginal costs are zero, $Cost=0$, then set the price where $\beta = -1$. If $Cost > 0$, then set the price where $\beta < -1$, the elastic region of the demand curve. This is easy to show:

Lower Bound on Monopoly Price

Claim: A profit-maximizing monopolist always selects P and Q in a region of the demand curve where $\beta < -1$

Proof: We can just use the monopoly markup rule from above to confirm this claim. If we cross-multiply the monopoly mark-up rule, we obtain:

$$Price = \left(\frac{\beta}{1+\beta} \right) Cost.$$

Since $Price \geq 0$ (i.e. you do not charge a negative price) and $Cost \geq 0$, then $\beta < -1$. If $\beta \geq -1$, then the formula above would imply setting a negative price.

This is a very useful result for a manager. If your team comes back to you with a price elasticity estimate $\beta \geq -1$, then you know that either you are underpricing your product OR your team made a mistake with their elasticity measurement.

Competitive Pricing and Elasticity

While it is called the monopoly pricing rule, this is really a misnomer. If the elasticity is measured in a market with multiple firms, this pricing rule tells a firm how to price assuming that the other firms don't react. Hence, what is missing from this pricing rule is competitor's reactions. If competitor's do not respond, then the monopoly markup rule may provide the optimal pricing strategy.

The monopoly pricing rule also assumes a firm selling one product. If a firm sells multiple products and there are cross-product effects, then the monopoly pricing rule ignores these effects. For example, in an electronics store the price of portable CD players may drive store traffic and increased store traffic leads to additional sales. In this case, the monopoly pricing rule is going to lead firms to overprice. Electronics stores may be willing to offer deep discounts on this type of item because they generate sales of other items.

In spite of these limitations of the monopoly markup rule, it is a very useful starting point for pricing and the assessment of *pricing power*.

Implied Elasticity

Let's consider the monopoly pricing rule again. Suppose a manager observes the current selling price and current costs. Can this information tell the manager anything useful about whether the price is too high or too low?

Using the same formula, rearrange terms and multiple the top and bottom of the right hand side by Quantity sold.

$$-\beta = \frac{\text{Price} \times \text{Quantity}}{(\text{Price} - \text{cost}) \times \text{Quantity}} = \frac{\text{Revenue}}{\text{Profits}}$$

Implied Elasticity

Hence the implied elasticity is given by total revenue divided by total profits.

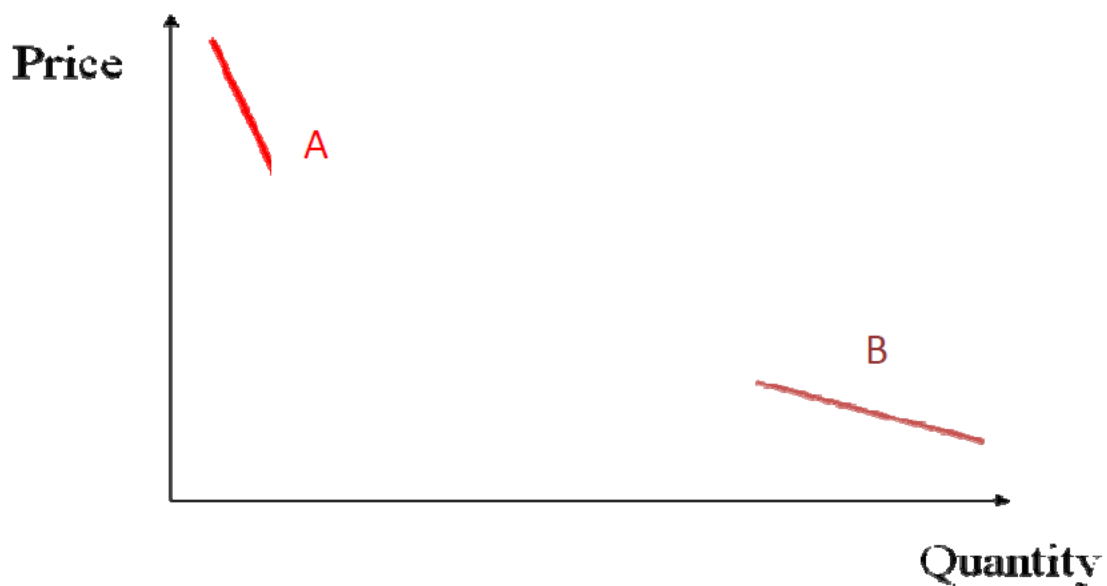
To illustrate, consider a simple example. Suppose that at the start of a season a firm generates \$300 revenue on the sale of ski boots and \$200 profit, the implied elasticity is (-1.5). At the end of the season, these same boots generate revenue of \$150 and profit of \$50. The implied elasticity is (-3).

This calculation provides a manager with a useful benchmark. For example, knowing that a typical elasticity is between -2 and -3, a manager could diagnose whether they are over or under pricing a product relative to this benchmark. A manager of a new, unique product with few substitutes would clearly want to be at or below the low end of this range (e.g. between -1 and -

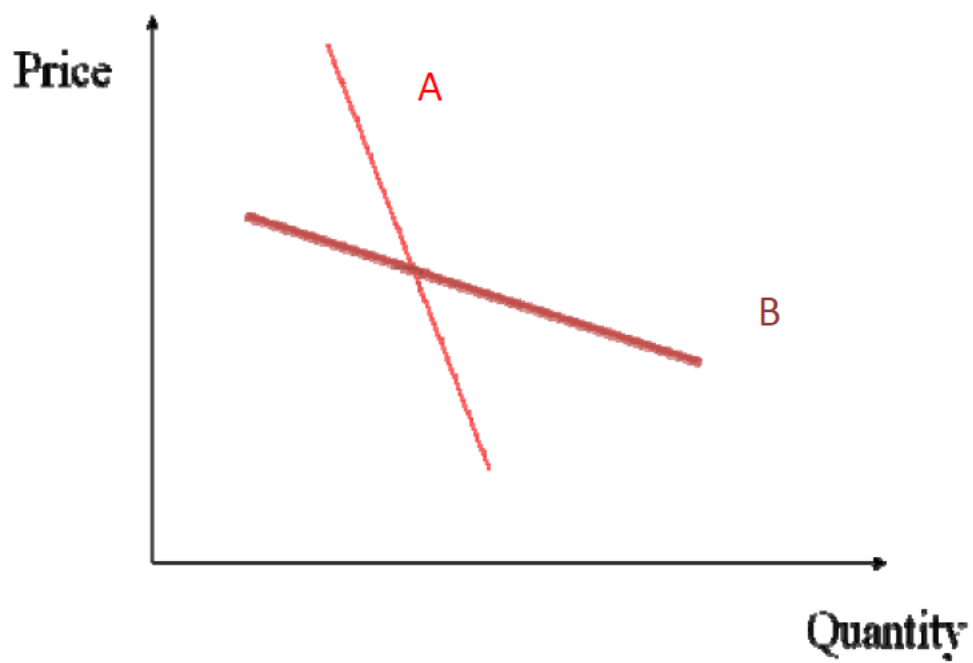
2.5). A manager of a highly competitive product may want to be sure to be at the other end of the scale (e.g. -2.5 or less). Knowledge (or estimates) of competitor's prices and profits could be used as a benchmark as well.

EXERCISES

1. Which Demand Curve is more elastic?

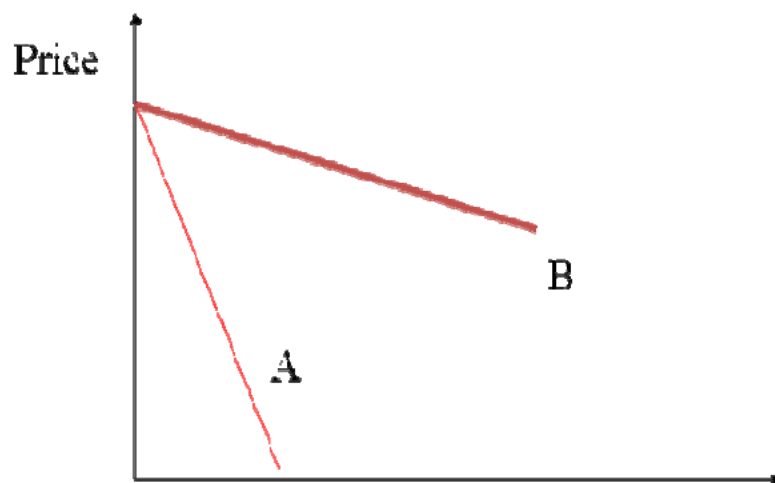


2. Which Demand Curve is more elastic?



Suppose Demand Curve A has slope: $dP/dQ = -10$ and demand curve B has slope $dP/dQ = -1$. Now compute the elasticities at the points ($P=10$, $q=2$) on A and B.

3. Which Demand Curve is more elastic?



Suppose Demand Curve A has slope: $dP/dQ = -10$ and demand curve B has slope $dP/dQ = -1$. Now compute the elasticities at the point where $q=2$ on A and B. Think about this one!

4. For the demand curve $Q = 100 - 5P$, what is the elasticity at $P = 10$? Will a profit maximizing firm ever set this price?
5. How should a manager use the monopoly pricing rule in setting price?

ESTIMATING THE PRICE-ELASTICITY OF DEMAND

Objectives

- To be able to estimate a price elasticity in a single product setting.
- To be able to estimate a price elasticity in a multi-product setting.

Model vs. Experiment

There are numerous ways to calculate a price elasticity. Two of the most common are to use either historical data or experimental data. When using historical data to calculate an elasticity, many things may co-vary with the change in price. For example, the marketing mix (promotion, coupon, advertising, etc.) might change. Another product's price might change or the set of competing products might vary. To capture all of these factors we need data. Further, we need a model that helps us isolate the effect of price while controlling for the impact of these other factors on sales. For example, if both in-store promotions and price change simultaneously, we need a model that describes how promotions and price relate to sales.

In an experiment, we try to control for as many factors as possible that co-vary. That is, we try to eliminate the need for a model by eliminating as many sources of variation. Analysis of experimental data often does not require a model. By controlling for other factors (i.e. holding other factors constant), one can focus on the changes in the variable of interest. Even if a model is needed to analyze the experiment, controlling for other factors reduces the complexity of the model. For example, if one was interested in suit sales from a catalog, many factors could affect demand for suits – even factors that might be hard to measure. However, by randomly assigning different versions of a catalog to customers, one could control for all these other factors. Hence, the only factors that one would expect to differ are those that are manipulated in the catalogs (e.g. different prices to different customers, different front cover of the catalog, etc.).

Since the analysis of experimental data is often straightforward, we will focus below on modeling historical field data.

Single Product Model

We estimate the price elasticity of demand using a *multiplicative* demand model.

$$Sales = \alpha \times Price^{\beta} \times Error$$

where we observe Sales and Price in our database, α and β are unknown parameters to be estimated, and Error is the statistical error of our model. We can estimate the model using linear regression by taking the log of both sides:

$$\ln(Sales) = \ln(\alpha) + \beta \ln(Price) + \varepsilon$$

where ε is just our regression error. What kind of data does one need to estimate the above model? Clearly, sales (in units) and prices are needed. One can also add other variables that

affect demand (e.g. time of year, advertising, temperature, etc.). Keep in mind that this is just like any other regression of the form $y = bx + \varepsilon$. Here, the y 's are $\ln(\text{Sales})$ and the x 's include $\ln(\text{Price})$ as well as other marketing variables.

Suppose we run the above regression and estimate the intercept, $\ln(\alpha)$, and the coefficient for $\ln(\text{Price})$, β . What is the price-elasticity of demand? From the note on elasticity, we know that by definition, the elasticity is the percent change in quantity divided by the percent change in price. A log-log model effectively takes the variables and expresses them as percentages. If we were to plot $\ln(\text{Sales})$ vs. $\ln(\text{Price})$, this would be identical to plotting the %change in sales against the % change in price. Hence, the slope of this regression line is the elasticity.

You could just as easily show that β is the elasticity formally by applying calculus to the multiplicative demand model, above, and computing: $\beta = \frac{dQ}{dP} \frac{P}{Q}$.

Pooling Data from Multiple Products

To estimate the above model, we assume that one has multiple data points for the same product. However, in many practical applications one may have many products and a small number of observations per product. One could simply estimate the multiplicative demand model with only a small number of observations per product.

Alternatively, one may be able to pool the data. The concern with pooling the data is that each product has a different demand curve. To control for this, one needs a benchmark for each product. The benchmark is used as a reference point, and changes in sales and price are all expressed relative to this benchmark. Note that each product should have its own benchmark. For example, suppose one had 100 products but only two observations per product. One observation came from New York City and the other observation came from LA. One could calculate the price-elasticity of demand for each product using the formula: Elasticity = (% change in quantity)/(% change in price). Alternatively, one could pool the data and estimate an elasticity for all the products. Let's see how.

We will now adapt the multiplicative (or log-log) model discussed above to accommodate "product effects". Let's denote the geographic market by i (i.e. NYC versus LA) and the product by j . The model is now:

$$\text{Sales}_{i,j} = \alpha_i \alpha_j \times \text{Price}_{i,j}^{\beta} \times \text{Error}$$

where α_i is the effect of being in market i , and α_j is the effect of product j . As before, we take logs on both sides of the equation to get the log-linear form:

$$\ln(\text{Sales}_{i,j}) = \ln(\alpha_i) + \ln(\alpha_j) + \beta \ln(\text{Price}_{i,j}) + \varepsilon_{i,j}$$

Suppose we use NYC as a benchmark. We now represent all the data relative to NY. To do this, we take the equation above and we subtract NY from LA. So we begin with

$$\begin{aligned} \ln(Sales_{NYC,j}) &= \ln(\alpha_{NYC}) + \ln(\alpha_j) + \beta \ln(Price_{NYC,j}) + \varepsilon_{NYC,j} \\ \ln(Sales_{LA,j}) &= \ln(\alpha_{LA}) + \ln(\alpha_j) + \beta \ln(Price_{LA,j}) + \varepsilon_{LA,j} \end{aligned}$$

And subtracting the two we get:

$$\ln\left(\frac{Sales_{LA,j}}{Sales_{NYC,j}}\right) = \ln\left(\frac{\alpha_{LA}}{\alpha_{NYC}}\right) + \beta \ln\left(\frac{Price_{LA,j}}{Price_{NYC,j}}\right) + \varepsilon_j$$

Notice how each variable is now expressed relative to NYC. If one had other geographies, one could subtract each geography from NYC (i.e. continue to use NYC as the benchmark). Note that the “ratios” are just the result of the property of logarithms: $\ln(X) - \ln(Y) = \ln(X/Y)$.

The intercept in the “differenced” regression model tells one about the relative size of the intercept in NY vs. LA. Notice that we cannot recover the absolute intercepts, α_{NYC} and α_{LA} , only the relative intercept, $\frac{\alpha_{LA}}{\alpha_{NYC}}$.

Suppose we have 100 products and estimate one common elasticity for all the products. Note how this differs from an alternative method of calculating a separate elasticity for each product. How should one interpret the single elasticity? Just like in the single product regression case, it is an average elasticity. If one suspects that the elasticity systematically varies by products, what would you do? The simplest method is to add dummy variables that interact product and price. For example, if the 100 products consist of electronics and furniture, we would write our demand model as follows:

$$Sales_{i,j} = \alpha_i \alpha_j \times Price_{i,j}^{\beta + \gamma D_{furniture,j}} \times Error_{i,j}$$

where $D_{furniture,j} = 1$ if product j is from the furniture category and $D_{furniture,j} = 0$ if not (i.e. j is from the electronics category). We would take logs and differences (as above) to estimate the model:

$$\ln\left(\frac{Sales_{LA,j}}{Sales_{NYC,j}}\right) = \ln\left(\frac{\alpha_{LA}}{\alpha_{NYC}}\right) + \beta \ln\left(\frac{Price_{LA,j}}{Price_{NYC,j}}\right) + \gamma \ln\left(\frac{Price_{LA,j}}{Price_{NYC,j}}\right) D_{furniture,j} + \varepsilon_j$$

If the elasticity of Furniture is different from the elasticity of Electronics, this will be picked up in coefficient γ . The price-elasticity of demand for furniture items is $(\beta + \gamma)$. For electronics, the price-elasticity of demand is β . The t-statistics from the regression would indicate whether the price-elasticity of furniture demand is different from the price-elasticity of electronics demand.

The main reason to pool the data rather than to calculate an elasticity for each product is that the

regression model uses all the data as efficiently as possible. Other methods will throw away information and hence yield a less efficient estimate of the elasticity.

Interpreting an Elasticity

As a manager, you may be handed an elasticity estimate rather than actually doing the calculation. The danger you should be aware of is GIGO (garbage in, garbage out). Before you can use an elasticity for decision making, you really need to understand how the elasticity was calculated. For example, you may be told that based on historical data demand is very inelastic (-0.3). A naïve manager might respond to this estimate by immediately raising the price. However, the astute manager would ask a few more questions such as:

- What kind of data were used to estimate the elasticity?
- If historical data was used, is it cross sectional, time series, or both?
- How much variation was there in price? What is the min/max price?
- How did our competitor's react to these price changes? Was this in the estimated elasticity?
- When prices varied, how was this communicated to customers?

These are just a few questions the astute manager would ask. The point is that calculating may be relatively easy. The hard part is assessing what you have calculated and how it can be used for decision-making.

Using Individual Purchase Data

The previous discussion focused on aggregate purchase data. In practice, firms typically track products at an aggregate level and not at the individual transaction level. However, with the development of more sophisticated technology firms now have the ability not only to collect but also to analyze individual transactions. For example, banks have maintained vast databases of customers and their individual transactions, supermarkets now actively collect individual purchases of their customers, and direct marketers collect individual responses to customized offers. In these settings, one can analyze individual data rather than aggregate data. The analysis of customer-level field data is mainly beyond the scope of this course. However, the data-driven marketing course devotes considerable time to the analysis of customer-level databases.